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History of Astronomy: Ancient Chinese Astronomy

The Celestial Empire

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Introduction

Of all the world's astronomical traditions, the Chinese one is the longest single story: a court-backed, office-held, archive-kept enterprise of watching the sky that ran, with changes of dynasty but not of purpose, for well over two thousand years, and that left behind the largest observational record any pre-modern civilization produced. The imperial astronomers of China logged eclipses, guest stars, comets, planetary conjunctions, and sunspots because the sky was the government's business: the emperor ruled by a mandate read in the heavens, and a flare, a blot, or a vanished sun was state information first and science second. This book is about that tradition, its instruments, its records, and the strange afterlife of those records, which modern astronomers now mine for data no telescope can recover: the exact appearances of comets two thousand years gone, the supernovae whose glowing remnants still broadcast, the eclipse timings that measure how the Earth's own spin has slowed.

What follows runs in the order this series keeps. The opening chapters define the tradition and its machinery: the mandate that put the sky on the official payroll, the cosmological theories argued in its literature, the instruments, gnomons, armillary spheres, water clocks, and the great surveys that mapped more than a thousand stars a thousand years before Europe printed its first charts. The middle chapters carry the record itself: the oracle-bone eclipse, the executed astronomers of legend, the star catalogs of the classical age, the Dunhuang chart, the guest stars of 1006 and 1054, the comet that Halley's calculation would later catch, and the calendar reforms that reached the Gregorian year's precision three centuries early. The later chapters describe the modern harvest, the use of the archive by present-day astronomy, and the tradition's own continuities into the present, when a nation that once forbade private star study operates some of the largest instruments on Earth.

A note on the record's character, because it explains the whole book: the Chinese observations were made without telescopes, but they were made by professionals, on schedule, with instruments of the state, and written into official histories that survived their dynasties. That combination, professional, continuous, and archived, is unique among the ancient traditions, and it is why a modern astronomer hunting the date of a medieval supernova starts, not with a European chronicle but with the annals of the imperial Chinese court.

A word on names and dating conventions, before the account begins. The tradition's persons are given in the romanizations standard in the modern literature, the astronomer-officials, the instrument-makers, the calendar reformers, and its documents by their usual English titles, with dynasties and reigns converted to the common era's calendar wherever the record permits. Dates before the common era are marked as such, and the tradition's own dating, by reign year and month, is explained where it matters, because the conversion of reign years to the continuous calendar is itself a scholarly tool, and the reader who understands it can date any entry in the archive. Volatile identifications, the contested star names, the disputed eclipse records, the legendary early material, are flagged as such in the chapters and checked in the fact notes; the tradition's firm results, and they are many, are stated plainly.

The record's scale deserves the opening note, because it is the tradition's defining fact and the reader should hold the numbers from the start. The court sky-watch kept its registers for roughly two thousand years of continuous office, from the bronze inscriptions of the Shang kings to the last memoirs of the late imperial bureau; the surviving annals preserve many hundreds of eclipse reports, dozens of documented calendar reforms, and the world's richest pre-telescopic files on comets, meteors, and the new stars that flare and fade; the volume's promise is that every chapter will keep this archive in view, because the tradition's story is, finally, the story of the longest running observational program any civilization has ever maintained.

The volume's map deserves a line too, because a two-thousand-year archive needs a route through it. The chapters run in the series' order, from the tradition's origins and its logic to its instruments, its records, and its modern afterlives, and the reader should know that the archive's sheer bulk forces a sampling rule: the volume quotes the standing exhibits, the eclipse that failed a dynasty, the guest star that became a nebula, the calendar that named the seasons for the world's largest state, and leaves the daybooks' thousand anonymous entries to stand for themselves; the map's honesty is the tradition's own method, the representative record kept for the whole, exactly as the imperial bureaus kept their digests for the dailies.

Chapter 1: The Discovery of Ancient Chinese Astronomy

The tradition's origins stand at the edge between legend and record, and the founding story is itself about the sky's seriousness. The ancient tale, told in the early classic of documents, describes the astronomers Xi and He, who

drank and neglected their duties and failed to predict a solar eclipse in the mythical reign, and were executed for it: whatever the event's historicity, the story's point was institutional, that the sky-watch was a state office whose failure was capital, and the theme repeats through the dynasties, eclipse forecasts as a test of the court's competence and the dynasty's mandate. The firmer origins are archaeological: the oracle bones of the Shang court, cattle and turtle shells inscribed for divination more than three thousand years ago, carry among their questions records of eclipses and of an occasional bright new star, the earliest datable astronomical observations from East Asia, made by scribes whose job, again, was the court's business of knowing what the sky meant.

The classical age, the centuries of the philosophers and the early empires, produced the tradition's working structure. The star catalogs attributed to the masters Gan and Shi of the fourth century prior to the common era enumerate the constellations and their stars by the hundreds; the historian Sima Qian, around the turn of that era, compiled the sky's official geography into a treatise that fixed the named asterisms, their meanings, and their portents; the court of the Han emperors established the pattern that would run for two millennia, a bureau of astronomy with official observers, official instruments, and an official duty to report everything unusual. By the turn of the common era the sky was mapped in a coordinate system of the twenty-eight lunar mansions, the sky divided into stations along the celestial equator through which the Moon and planets passed night by night, and the calendar, the tradition's central product, was being tuned to the eclipse cycle by court mathematicians.

The tradition's discovery by the West, which is a different story, waited for the modern era. Jesuit missionaries at the Ming and Qing courts, from the seventeenth century onward, found a functioning imperial observatory, staffed by hereditary astronomers, holding records whose depth their own science could not match; the eclipse predictions were handed to the newcomers to compute, a transfer of technique that began the tradition's eclipse into European method, while the archive itself stayed in Chinese hands and Chinese script. The full value of the records emerged in the twentieth century, when astronomers hunting the past, for comets, for supernovae, for the Earth's slowing spin, learned to read the annals, and found in them the observational spine of half the world's pre-telescopic history, a discovery that is still, a century on, producing papers.

The chapters that follow take the tradition in the order of this outline: the offices and the instruments, the theories and the calendars, the records and the finds, and the modern mining of the whole. The discovery at the base of it all is the one the oracle-bone scribes made without knowing it: that an observation written down for the state outlives the state, and that the sky, properly archived, is a time machine.

The silk and bamboo detour belongs in the discovery chapter, because the classical record kept surprises for the modern era. The excavated texts of the last half-century, the Han-era tomb libraries, the manuscript caches, have added astronomical material the transmitted histories never held: star tables on silk from the second century before the common era, eclipse and planetary records on bamboo slips, divination texts with sky lore embedded, the finds having pushed the tradition's documented precision earlier with each excavation. The archaeological finds also settled, in the scholars' favor, a long-standing doubt: the transmitted annals, checked against the excavated documents where they overlap, prove to be honest copies more often than not, the compilers having preserved their sources with a fidelity that lets the modern reader trust the archive's spine. The discovery of ancient Chinese astronomy, in the philological sense, is therefore still in progress, the tradition's earliest chapters being recovered from the ground even as its later ones are being digitized.

The origins chapter should add the archaeological anchor, because the tradition's depth is now a matter of excavation rather than legend. The inscribed oracle bones of the late Shang court, asking the royal ancestors about sacrifices and harvests, carry dated sky reports among their questions: eclipses of the sun, an eclipse of the moon, and the names of stars, all incised a good three thousand years before the present; the find matters because it dates the office before the empire, the court's sky habit predating the unified state by a millennium, and the chapter's standing point is that the legend of the drunken astronomers deserves its archaeological footnote: whatever the tale's truth, the bones prove that the consequence of missing an eclipse was a matter of state record already in the bronze age.

The origins chapter should also carry the star-name stratum, because the sky's earliest Chinese vocabulary is itself a founding document. The names the tradition gave its stars are offices and palaces, wardens and generals, the celestial moats and gates of a mirrored government: the asterisms of the imperial enclosure around the pole, the four symbolic beasts holding the four seasonal quadrants, the mansions strung along the sky's great road; the chapter's dating note is that this vocabulary is old, appearing in the early poems and on the ancient tombs' decorated ceilings, so the sky-as-bureaucracy predates the bureaus that wrote the surviving files, and the tradition's founding, read in its names, is a theory of the state written in the stars before it was written in the annals.